



Assignment

Spring 2026 2nd Semester

Course Title: Circuit Analysis Lab

Course Code: CMPE-121L

Topic(s): Lab 6 Verification Of Kirchhoff's Laws

SUBMITTED TO:

Mr. Mehmood-ul-Haq

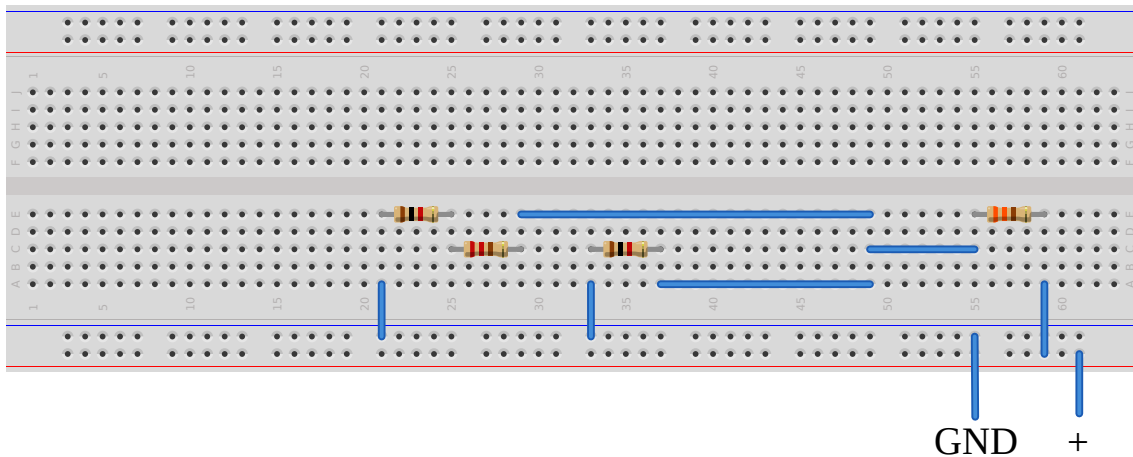
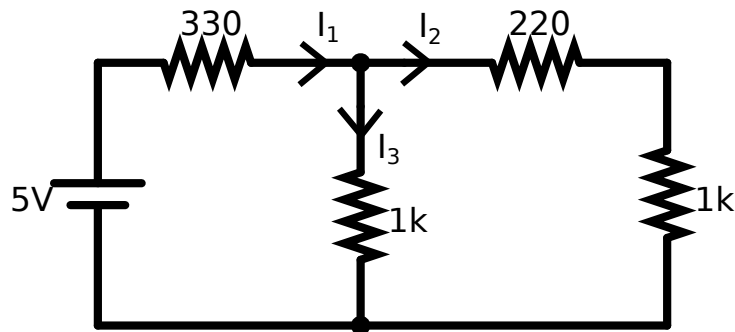
SUBMITTED BY:

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Submission Date: 27 Apr, 2026

Department of Computer Engineering
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Experiment 1



Breadboard circuit layout: No measurements being taken

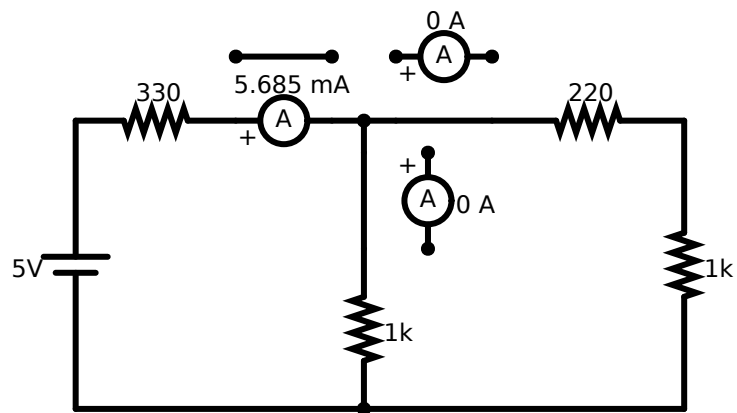
Practical Procedure

The objective of this experiment is to verify Kirchhoff's Current Law (KCL) by analyzing the nodal current distribution in a series-parallel resistive network. The circuit is constructed using a **Regulated Power Supply (RPS)** as the DC source, with $R_1 = 330\Omega$ connected in series to a junction that splits into two parallel paths. Branch A contains a $1\text{ k}\Omega$ resistor, while Branch B consists of a 220Ω and $1\text{ k}\Omega$ resistor in series. We take each current reading for different RPS voltages. For this experiment, we will supply 5V, 15V and 25V from our RPS.

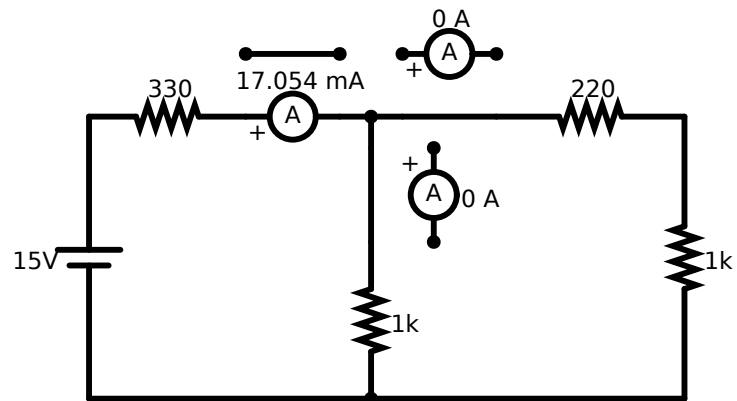
To measure the branch currents, the multimeter dial is set to the **20mA** or **200mA DC** range (indicated by the **A—** symbol). The black probe is connected to the

COM port while the red probe is inserted into the **mA μ A** terminal. Measuring current requires the DMM to be connected in **series**, which necessitates breaking the circuit at each specific node to allow the charge to flow through the meter. The total current (I_1) is measured at the output of the RPS, while I_2 and I_3 are captured by placing the meter in series with their respective parallel branches. This arrangement allows us to verify that the current entering the main junction is equal to the sum of the currents exiting through the parallel paths.

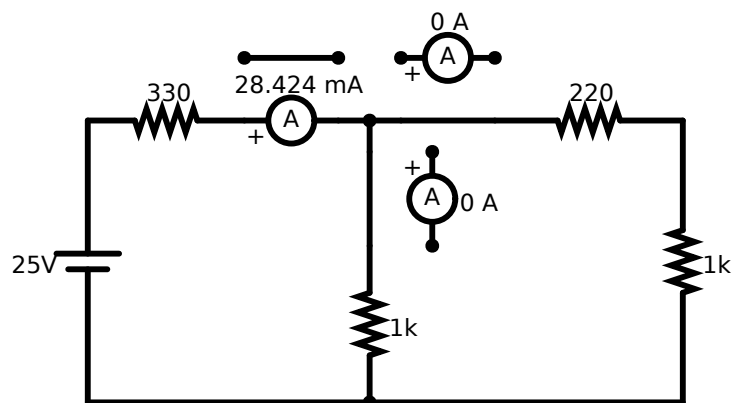
Theoretical predictions assume ideal conditions; however, minor discrepancies may be observed in simulated or physical environments. Factors such as the **burden voltage** of the ammeter and the **internal resistance** of the connecting leads of the multimeter and/or heat damage and corrosion on the breadboard contact points can introduce slight deviations.



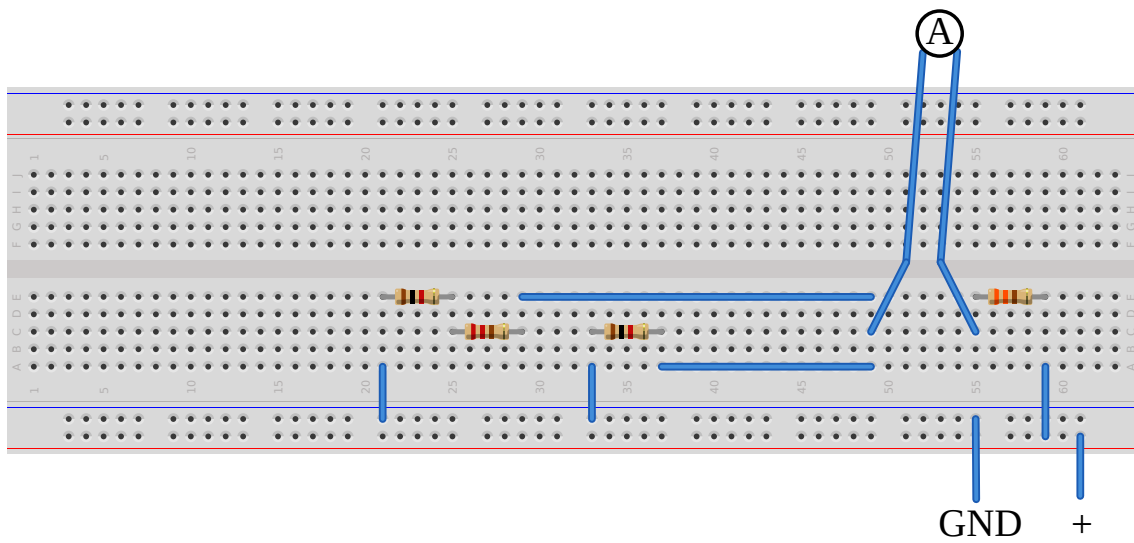
Falstad: Current I_1 due to 5V source



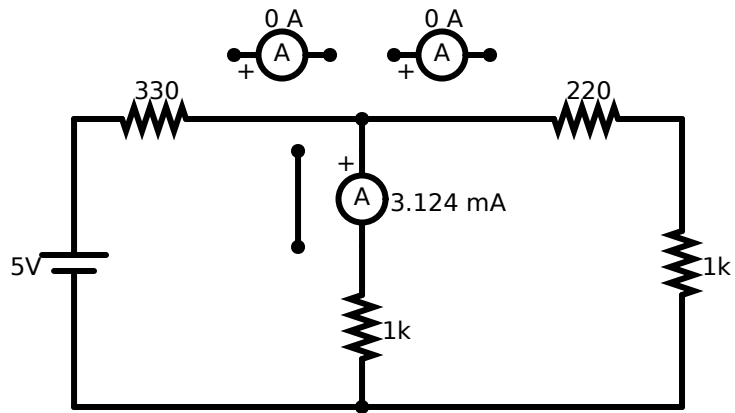
Falstad: Current I_1 due to 15V source



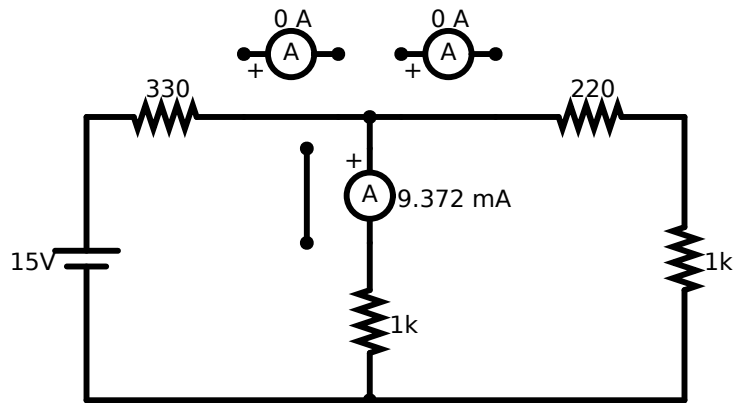
Falstad: Current I_1 due to 25V source



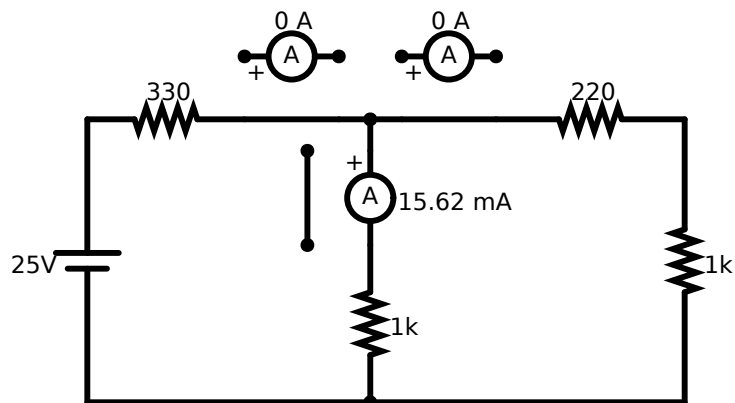
Breadboard circuit layout: Ammeter connection to measure I_1



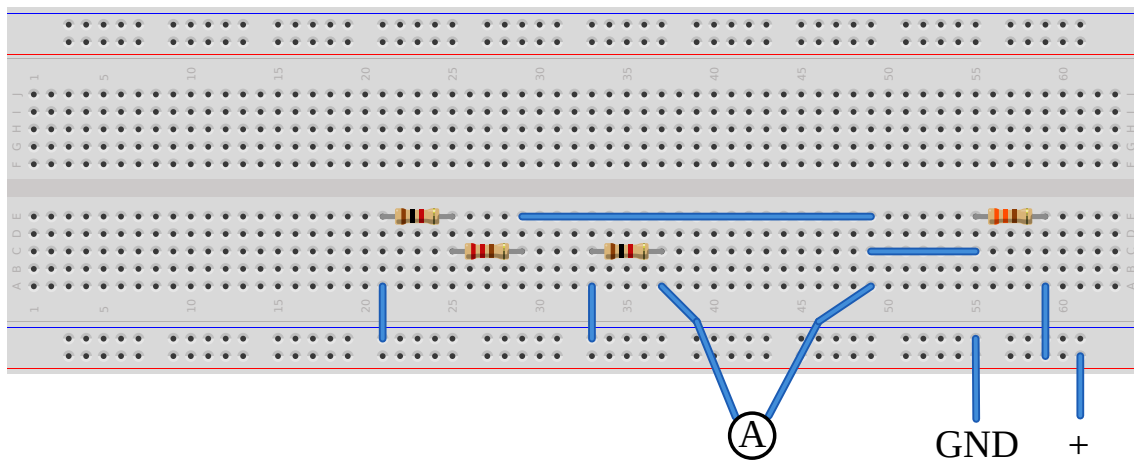
Falstad: Current I_2 due to 5V source



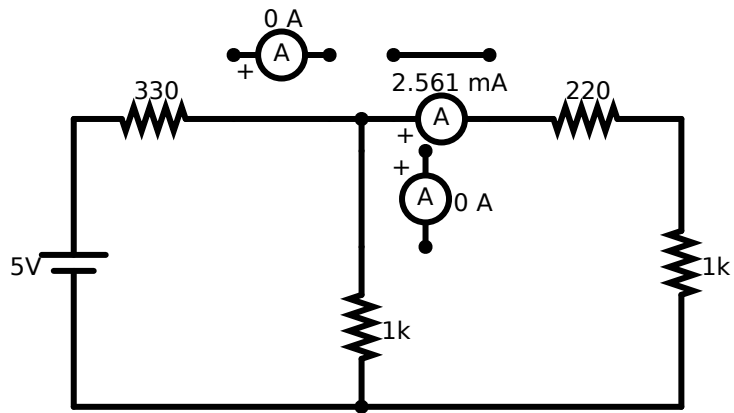
Falstad: Current I_2 due to 15V source



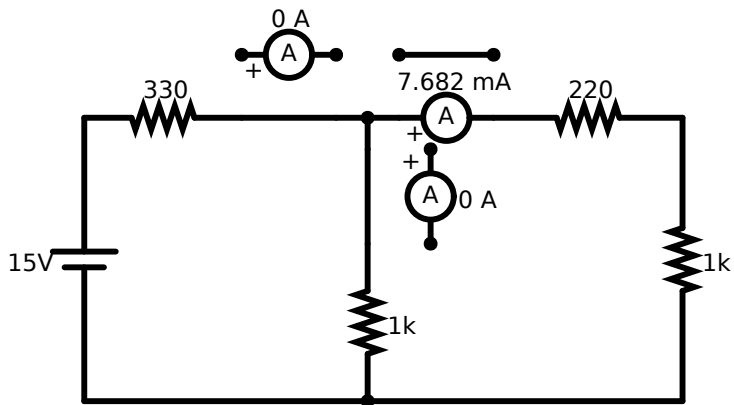
Falstad: Current I_2 due to 25V source



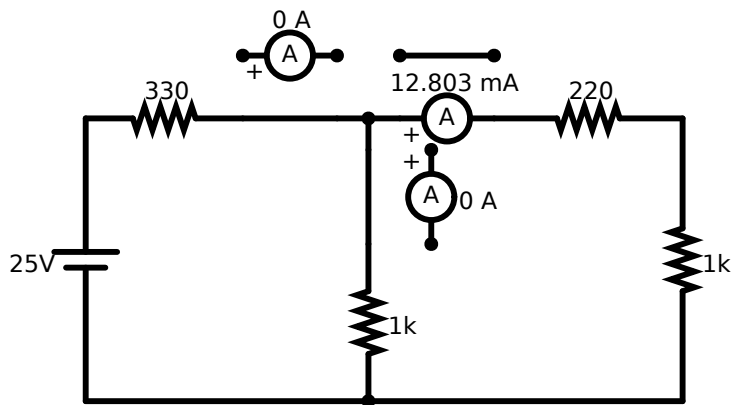
Breadboard circuit layout: Ammeter connection to measure I_2



Falstad: Current I_3 due to $5V$ source



Falstad: Current I_3 due to $15V$ source



Falstad: Current I_3 due to $25V$ source

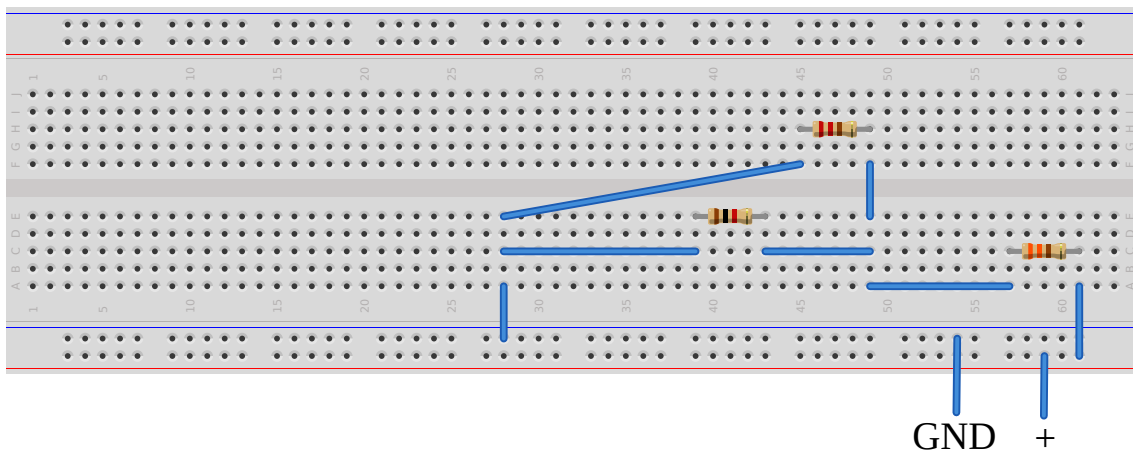
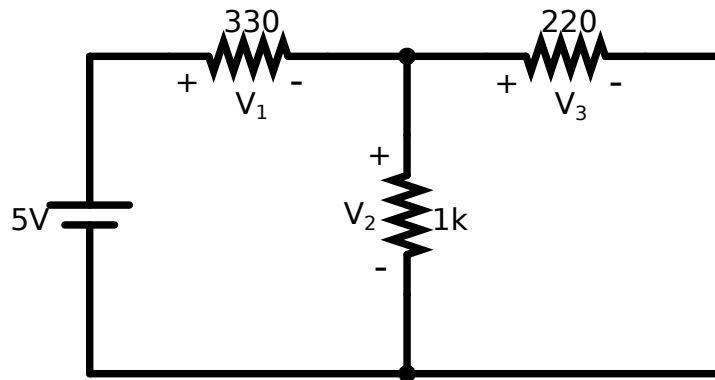
Breadboard circuit layout: Ammeter connection to measure I_3

Experiment 1 Data

Practical Values

Sl. No.	Voltage E	Current			$I_1 = I_2 + I_3$
	V	I_1	I_2	I_3	
		mA	mA	mA	mA
1	5	5.685	3.124	2.561	$5.685 = 5.685$
2	15	17.054	9.372	7.682	$17.054 = 17.054$
3	25	28.424	15.620	12.803	$28.424 \approx 28.423$

Experiment 2



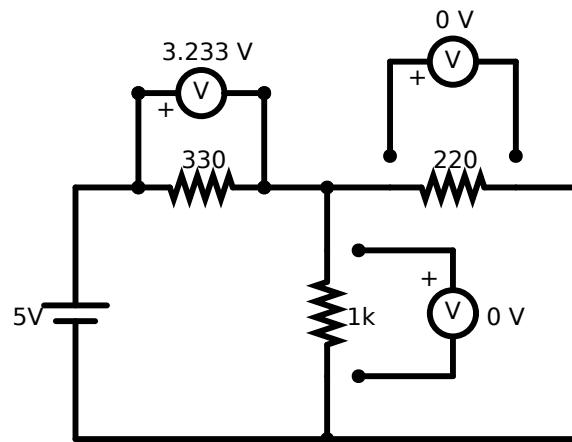
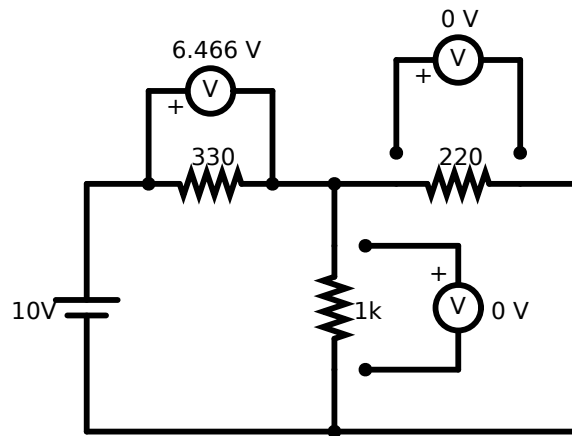
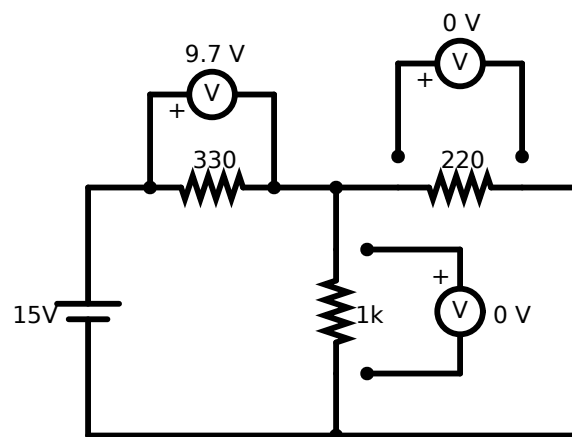
Breadboard circuit layout: No measurement being taken

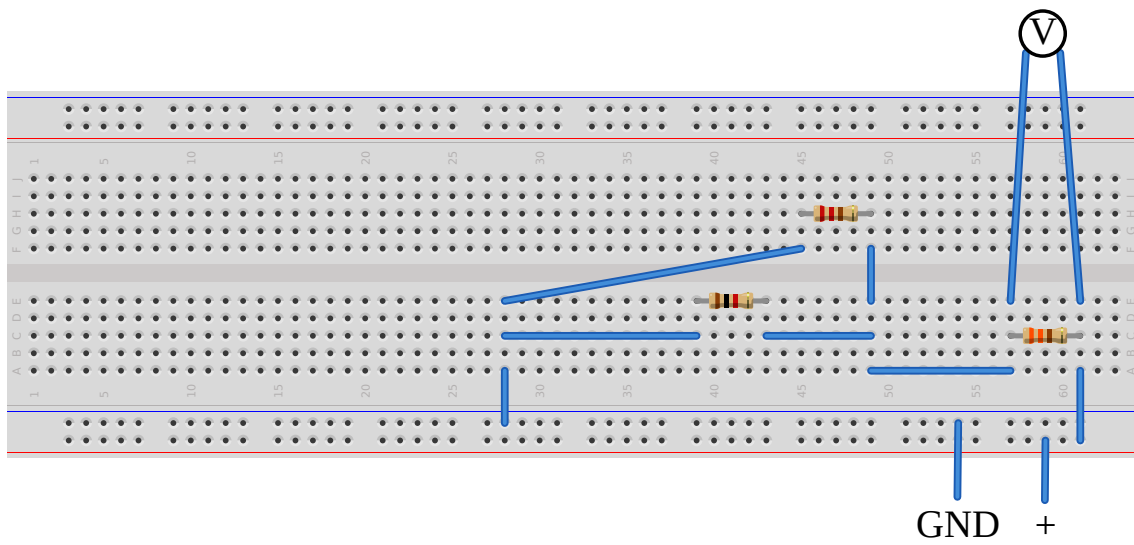
Practical Procedure

The objective of this experiment is to verify Kirchhoff's Voltage Law (KVL) by examining the distribution of potential differences across a series-parallel resistive loop. The network is assembled by connecting a 330Ω resistor in series with a parallel bank composed of a $1\text{ k}\Omega$ resistor and a 220Ω resistor. The circuit is driven by the **Regulated Power Supply**, and the actual source voltage (E_1) is monitored at increments of 5 V, 10 V, and 15 V to observe how the potential drops scale linearly across the different resistive elements.

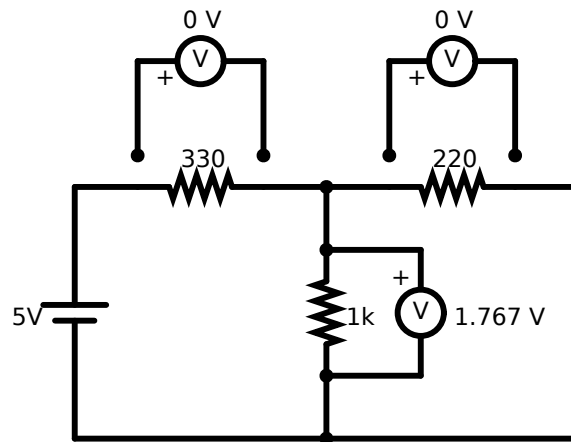
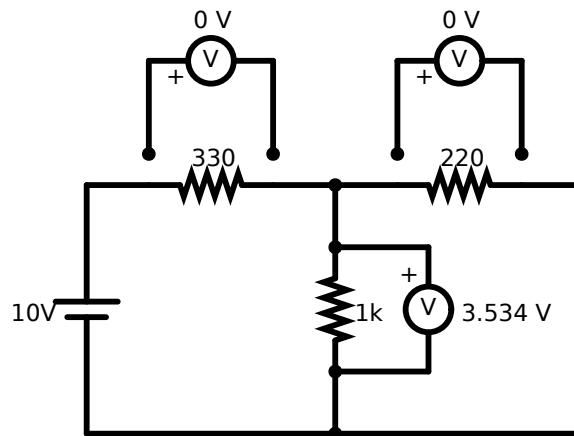
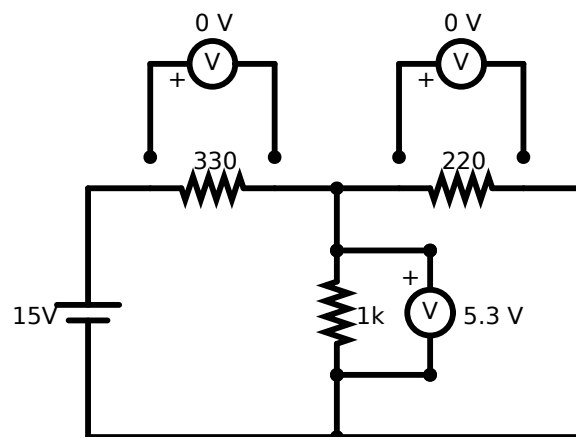
For voltage measurements, the multimeter dial is set to the **20V DC** range (indicated by the **V—** symbol). The red probe is inserted into the **VΩHz** terminal and the black probe into the **COM** port. To measure the potential drops V_1 , V_2 , and V_3 , the probes are placed in **parallel** across each resistor. Because V_2 and V_3 are measured across parallel branches, they share common nodes and should yield identical readings. KVL is verified by confirming that the sum of the voltage drop across the series resistor (V_1) and the voltage across the parallel bank (V_2) equals the total electromotive force provided by the RPS.

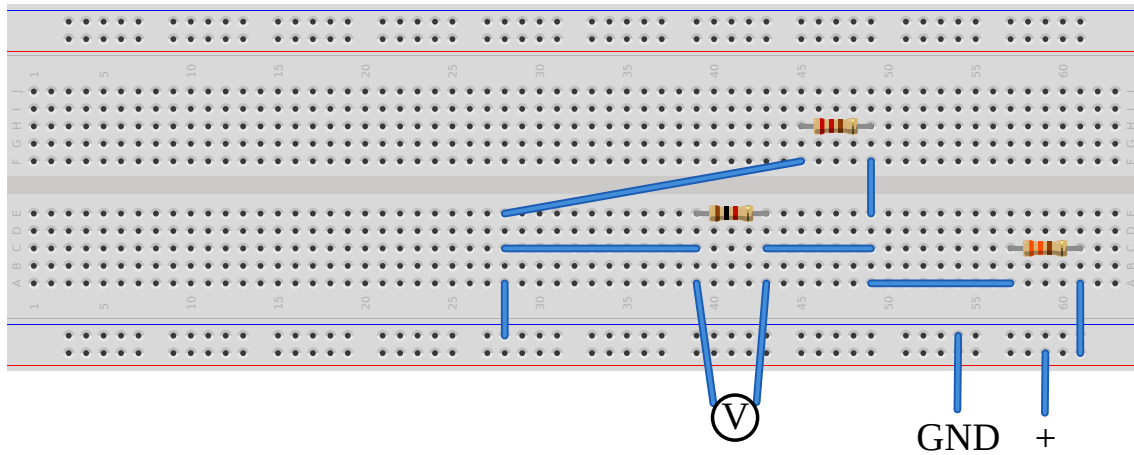
Discrepancies in KVL measurements typically arise from the **loading effect** of the voltmeter, which possesses a high but finite internal impedance (typically $10\text{ M}\Omega$). In a physical setup, contact resistance at the breadboard nodes and the precision of the power supply's voltage regulation can also influence the results.

Falstad: Voltage V_1 due to 5V sourceFalstad: Voltage V_1 due to 10V sourceFalstad: Voltage V_1 due to 15V source

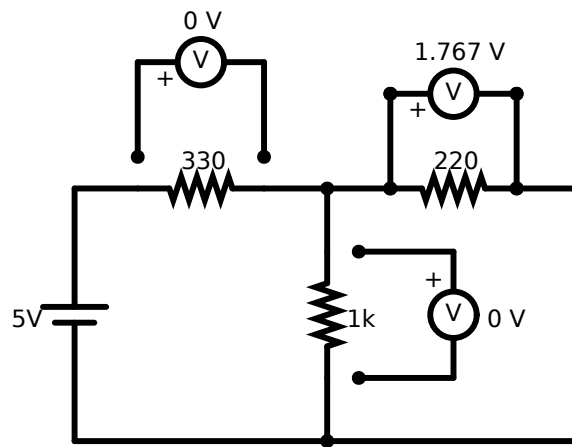
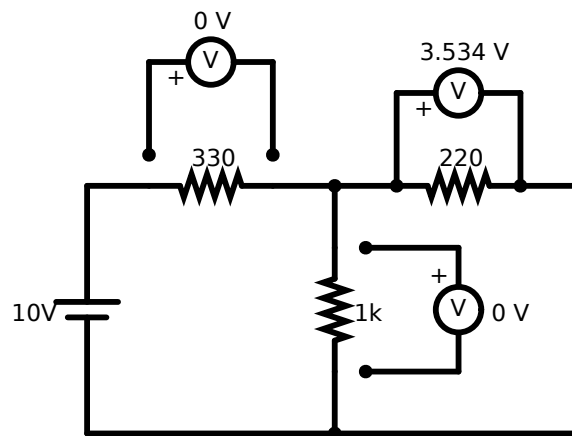
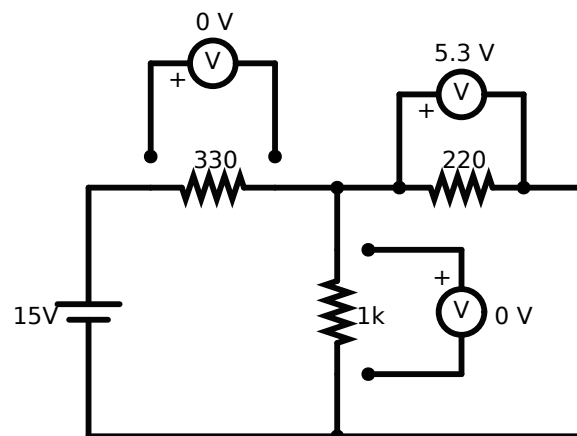


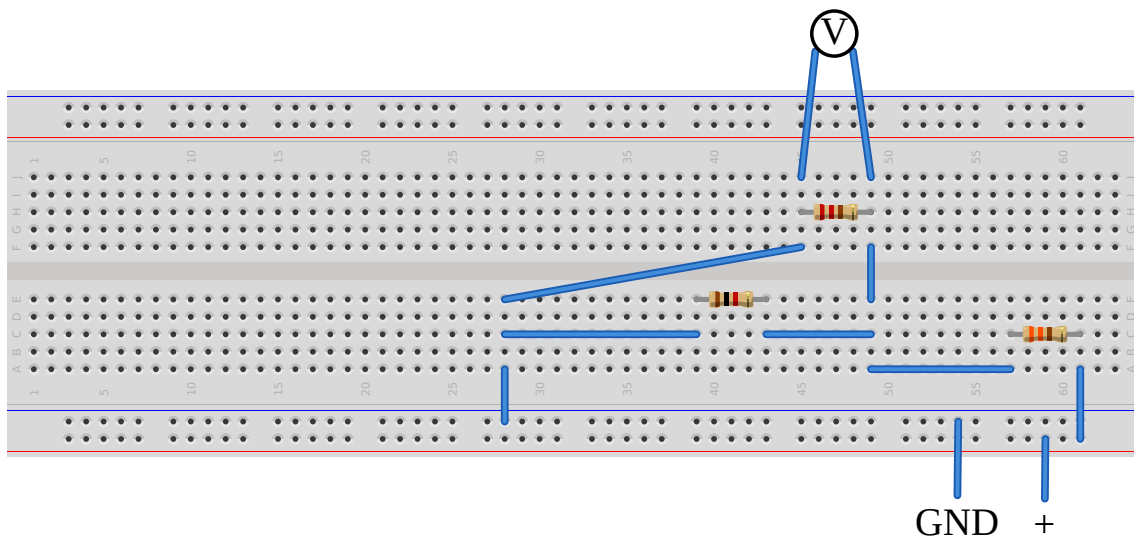
Breadboard circuit layout: Voltmeter connection to measure V_1

Falstad: Voltage V_2 due to 5V sourceFalstad: Voltage V_2 due to 10V sourceFalstad: Voltage V_2 due to 15V source



Breadboard circuit layout: Voltmeter connection to measure V_2

Falstad: Voltage V_3 due to 5V sourceFalstad: Voltage V_3 due to 10V sourceFalstad: Voltage V_3 due to 15V source



Breadboard circuit layout: Voltmeter connection to measure V_3

Experiment 2 Data

Practical Values

Sl. No.	RPS	Voltage Drops			$E_1 = V_1 + V_2$
	E_1	V_1	V_2	V_3	
	V	V	V	V	V
1	5	3.230	1.767	1.767	$5 \approx 4.997$
2	10	6.466	3.534	3.534	$10 = 10$
3	15	9.700	5.300	5.300	$15 = 15$

Post-Lab Questions

1. Define Kirchhoff's current law.

Kirchhoff's Current Law (KCL), also known as Kirchhoff's First Law, states that the algebraic sum of all currents meeting at a node or junction in an electrical circuit is zero. This implies that the total current entering a node must be exactly equal to the total current leaving that same node. It is a fundamental consequence of the Law of Conservation of Charge, assuming that charge does not accumulate at any single point within a steady-state circuit.

2. Define Kirchhoff's voltage law.

Kirchhoff's Voltage Law (KVL), or Kirchhoff's Second Law, states that the algebraic sum of all electrical potential differences (voltages) around any closed loop or mesh in a network is zero. In practical terms, this means that the sum of the electromotive forces (EMFs) supplied to the loop is equal to the sum of the potential drops (voltage across resistors) within that same loop. This law is an application of the Law of Conservation of Energy.

3. What are the applications of regulated power supply?

A Regulated Power Supply (RPS) is an essential instrument in electronic design and testing due to its ability to maintain a constant output voltage regardless of variations in load current or input voltage. Its primary applications include:

- Powering sensitive integrated circuits and microprocessors that require stable DC bias.
- Serving as a reference voltage source for laboratory experiments and circuit prototyping.
- Charging batteries with precise voltage and current control.
- Testing the tolerance and operational limits of electronic components under controlled conditions.

4. What are the limitations of KCL?

While KCL is a cornerstone of circuit analysis, it has specific limitations:

- **High-Frequency Limits:** KCL is only strictly applicable to "lumped" parameter circuits. At very high frequencies (RF/Microwave), where the wavelength of the signal is comparable to the physical dimensions of the circuit,

parasitic capacitance and radiation effects cause charge to "leak," making KCL appear inaccurate.

- **Varying Magnetic Fields:** If a node is subject to a rapidly changing external magnetic flux, it can induce currents that are not accounted for by simple nodal analysis.
- **Non-Steady State:** In cases where charge accumulation occurs (such as the interior of a capacitor plate), KCL must be modified to include displacement current (dD/dt).